ETH zürich

KOF

# Principles of Macroeconomics

**Lecture 13: Philips Curve / Supply-side policies**

Prof. Dr. Jan-Egbert Sturm

# Schedule

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# Remember: Natural Rate of Unemployment

- The ***natural rate of unemployment*** is unemployment that does not go away on its own even in the long run
- There are two reasons why there is a positive natural rate of unemployment:
  1. job search (frictional unemployment)
  2. wage rigidity (structural unemployment)
     - Minimum-wage laws
     - Unions
     - Efficiency wages

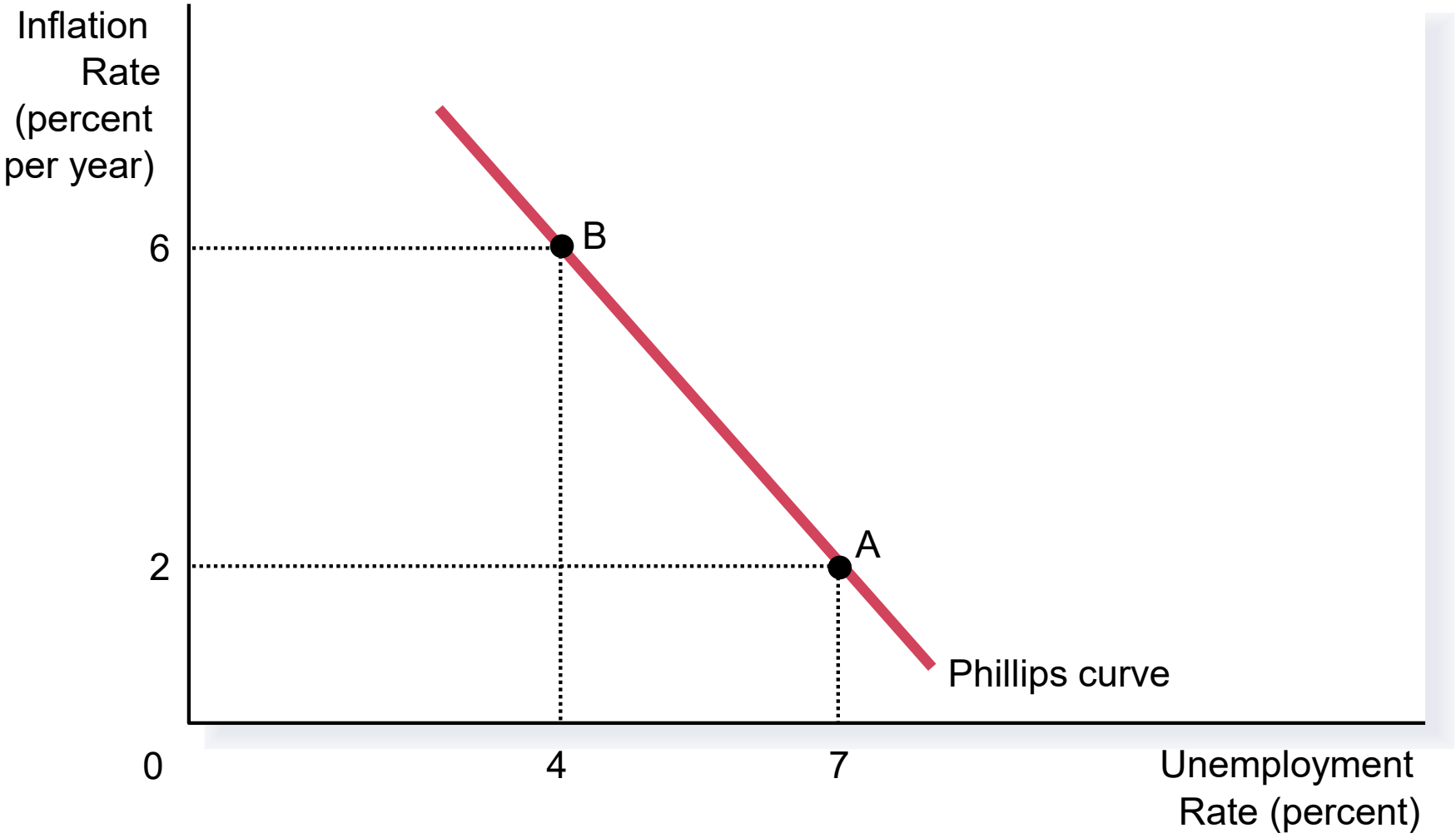
# Remember: The Quantity Equation & Classical Dichotomy

- The ***quantity equation***: $M \times V = P \times Y$ $\rightarrow \Delta M/M + \Delta V/V = \Delta P/P + \Delta Y/Y$
  - relates the quantity of money ($M$) to the nominal value of output ($P \times Y$)
  - It shows that an increase in money must be reflected in one of three other variables:
    - the price level must rise,
    - the quantity of output must rise, or
    - the velocity of money must fall
- The velocity of money is considered to be (relatively) fixed: $V = \bar{V}$
- The ***classical dichotomy***
  - Real economic variables do not change with changes in money: $Y = \bar{Y}$
- Hence, changes in money supply only affect the price level: $\Delta M/M = \Delta P/P$
  (as it is defined/assumed that $\Delta V = 0$ & $\Delta Y = 0$)

# Short-Run Trade-Off between Inflation and Unemployment: THE PHILLIPS CURVE

- Society faces a short-run trade-off between unemployment and inflation
- If policymakers expand aggregate demand, they can lower unemployment, but only at the cost of higher inflation
- If they contract aggregate demand, they can lower inflation, but at the cost of temporarily higher unemployment
- The Phillips curve shows the short-run trade-off between inflation and unemployment

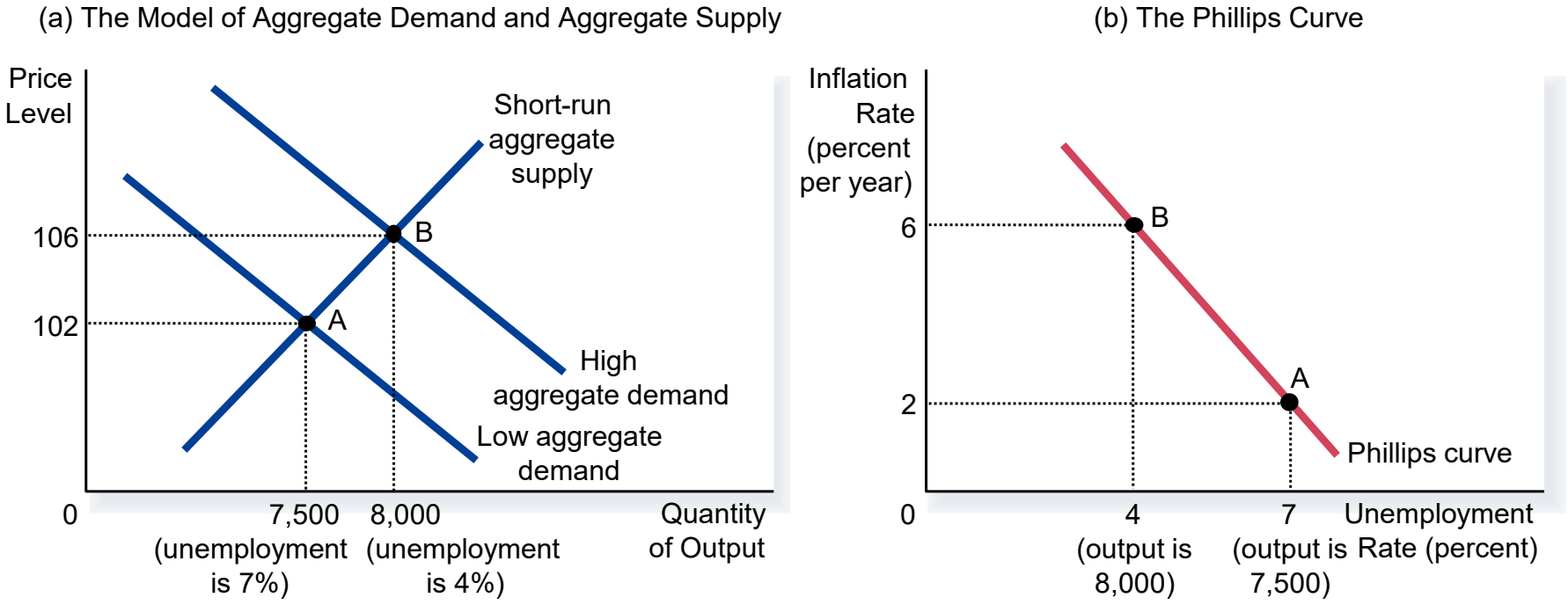
# The Phillips Curve



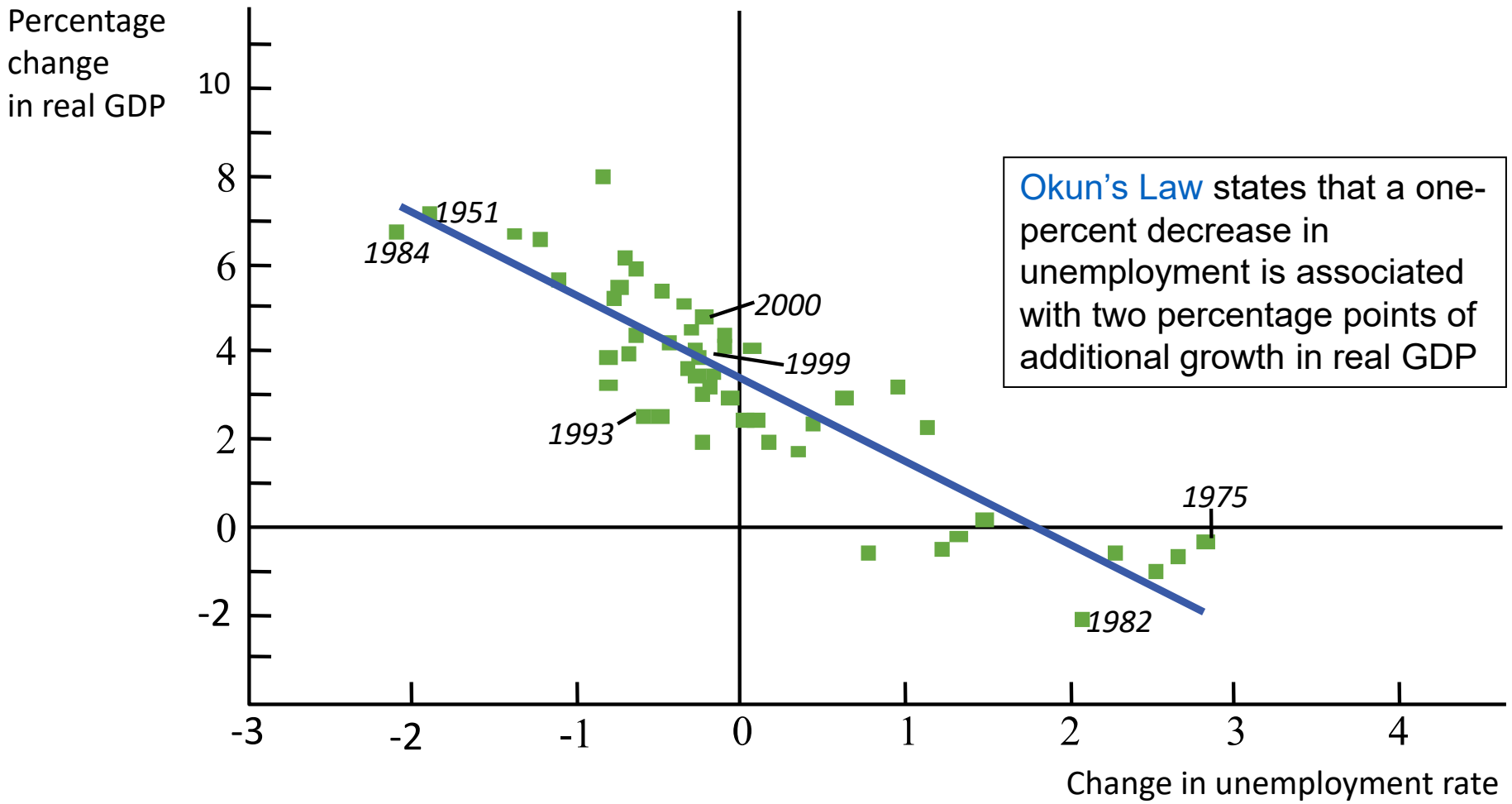
# Aggregate Demand, Aggregate Supply and the Phillips Curve

- The Phillips curve shows the short-run combinations of unemployment and inflation that arise as shifts in the aggregate demand curve move the economy along the short-run aggregate supply curve
- The greater the aggregate demand for goods and services, the greater is the economy's output, and the higher is the overall price level
- A higher level of output results in a lower level of unemployment

# How the Phillips Curve is Related to Aggregate Demand and Aggregate Supply

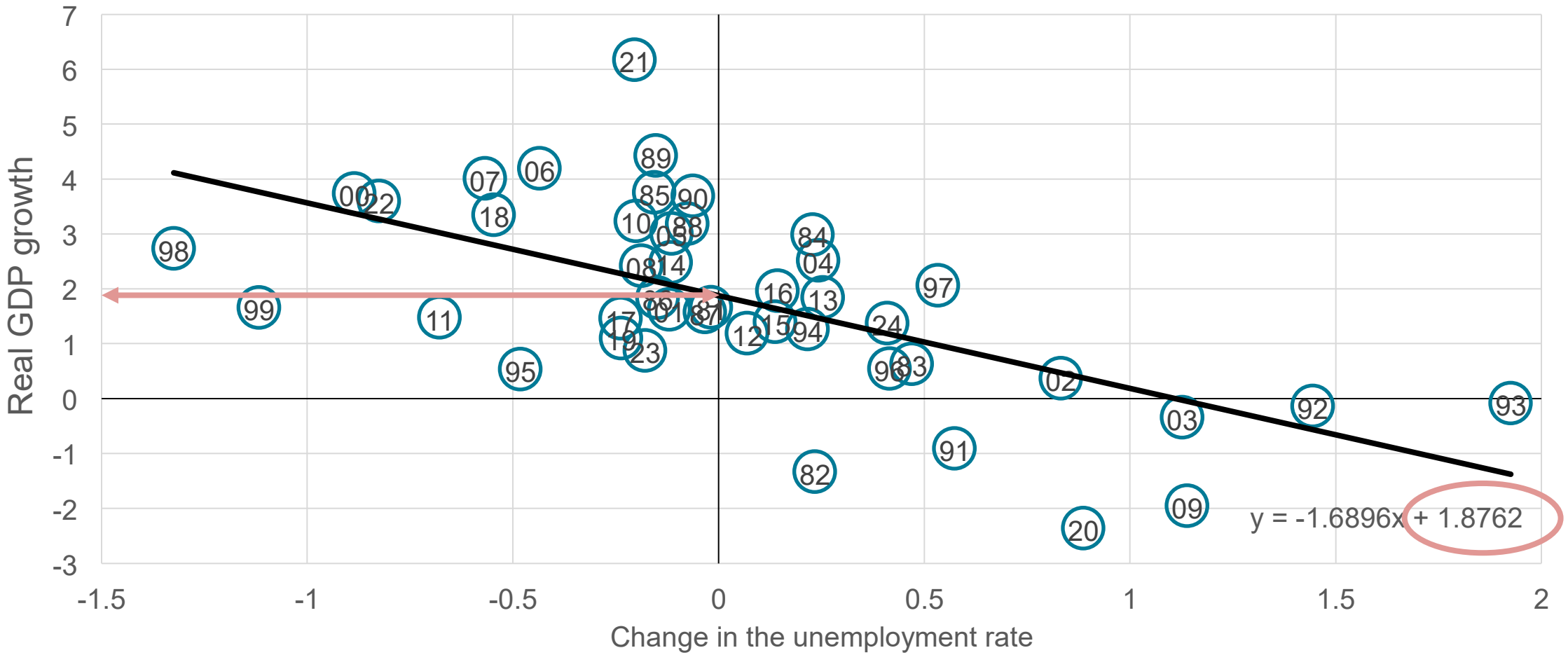


# Okun's Law



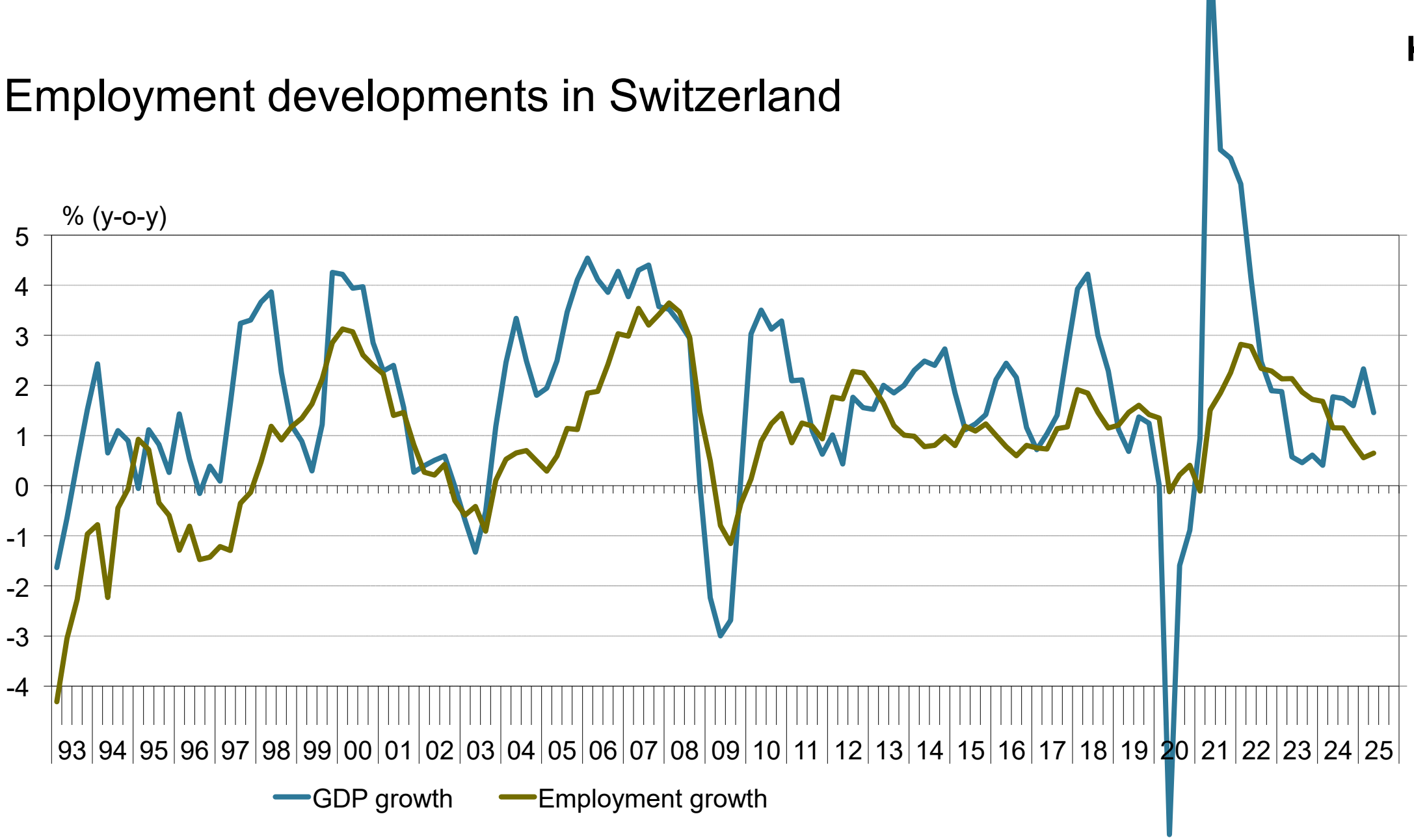

Okun's Law states that a one-percent decrease in unemployment is associated with two percentage points of additional growth in real GDP

# Okun's Law for Switzerland (1981-2024)



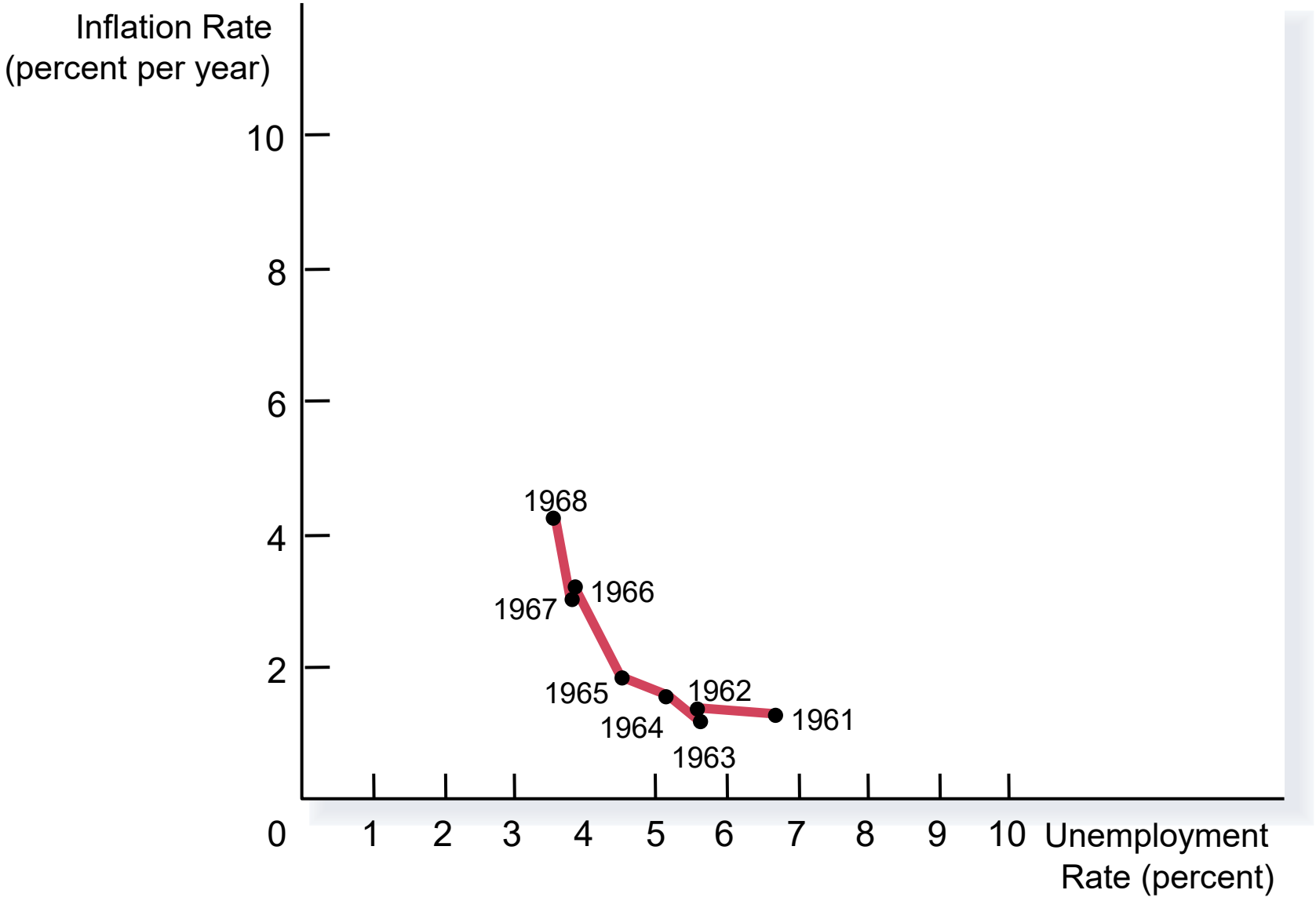

Sources: Seco, SFSO

# Employment developments in Switzerland




Sources: Seco, SFSO, KOF

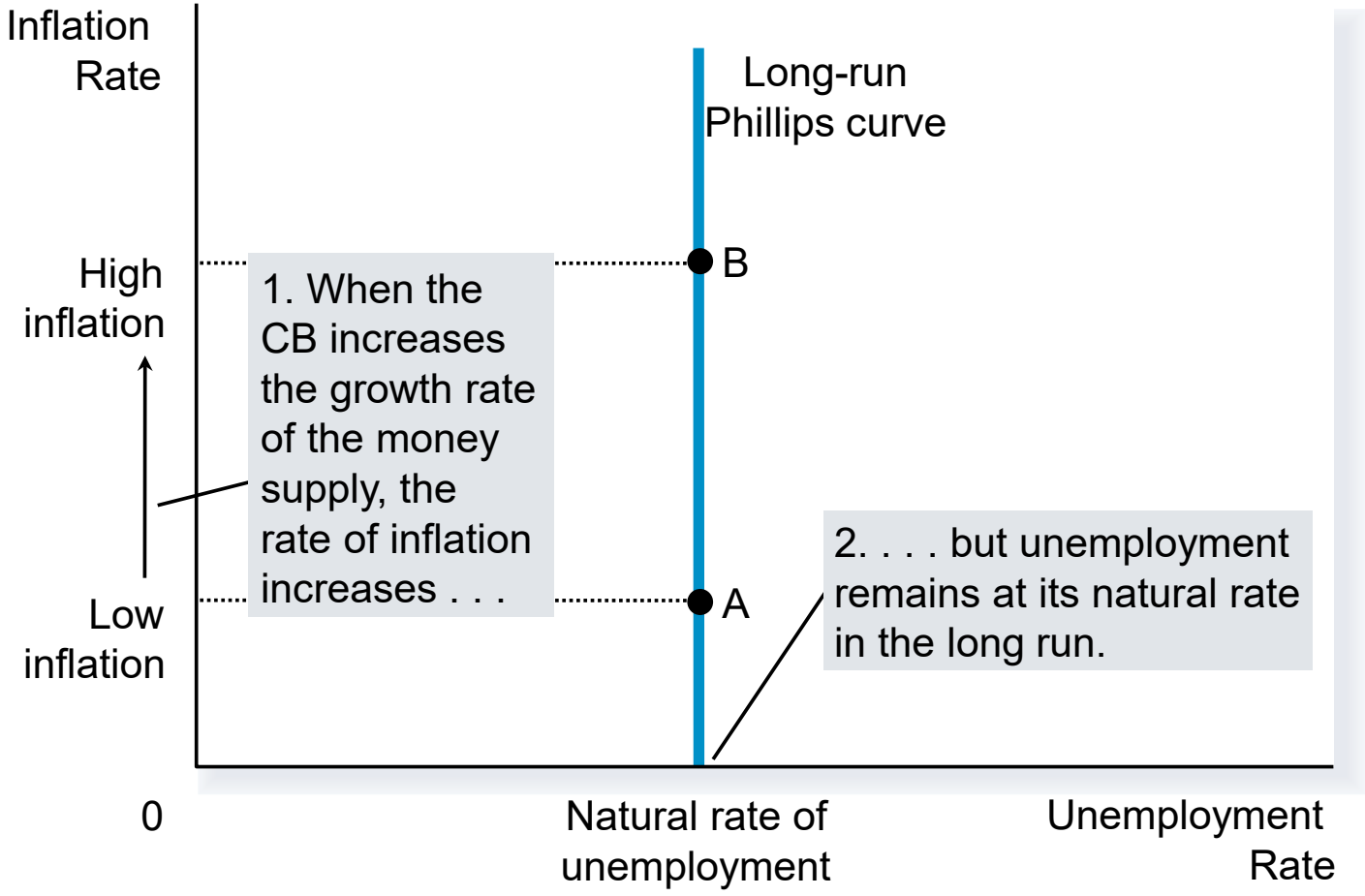
# The Phillips Curve in the 1960s (USA)



# The Long-Run Phillips Curve

- The (short-run) Phillips curve seems to offer policymakers a menu of possible inflation and unemployment outcomes

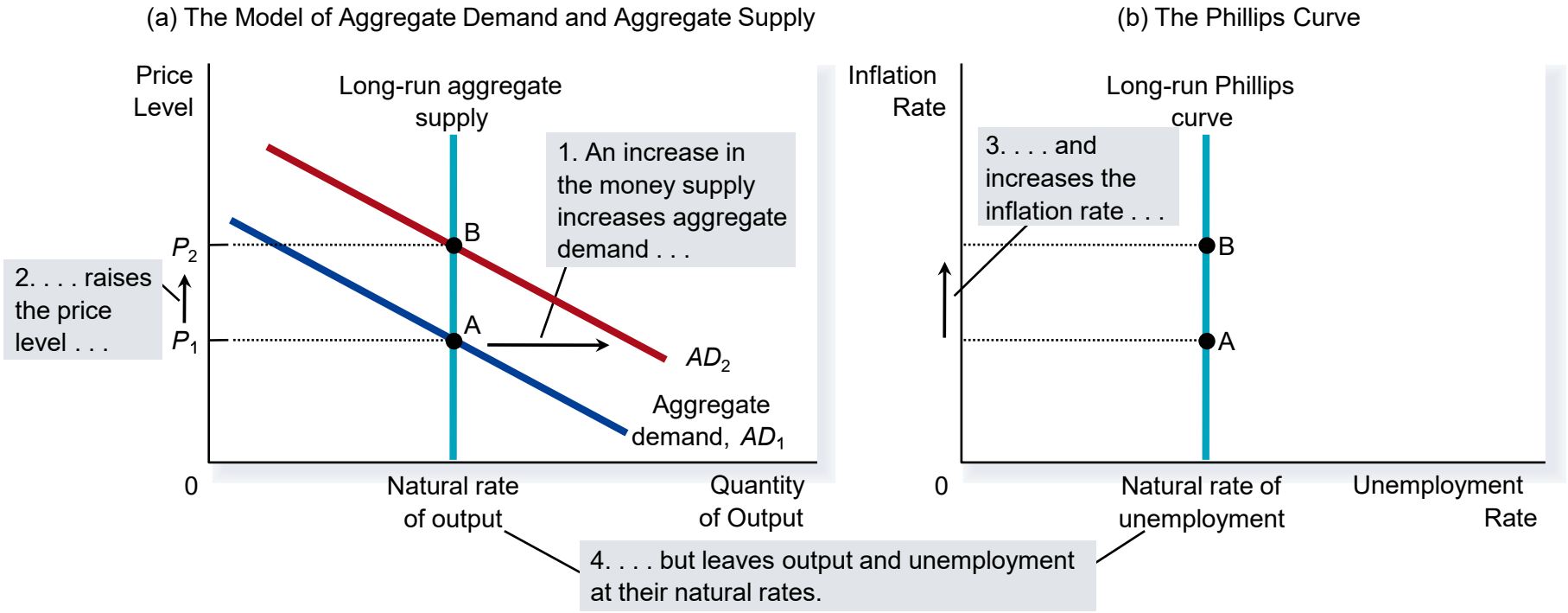
- In the 1960s, Friedman and Phelps concluded that inflation and unemployment are unrelated in the long run
  - As a result, the long-run Phillips curve is vertical at the natural rate of unemployment
  - Monetary policy could be effective in the short run but not in the long run



# The Meaning of “Natural”

- The “natural” rate of unemployment is the rate to which the economy gravitates in the long run
- The natural rate is not necessarily desirable, nor is it constant over time
- Monetary policy cannot change the natural rate,
  but other government policies that strengthen labour markets can

# How the Phillips Curve is Related to Aggregate Demand and Aggregate Supply



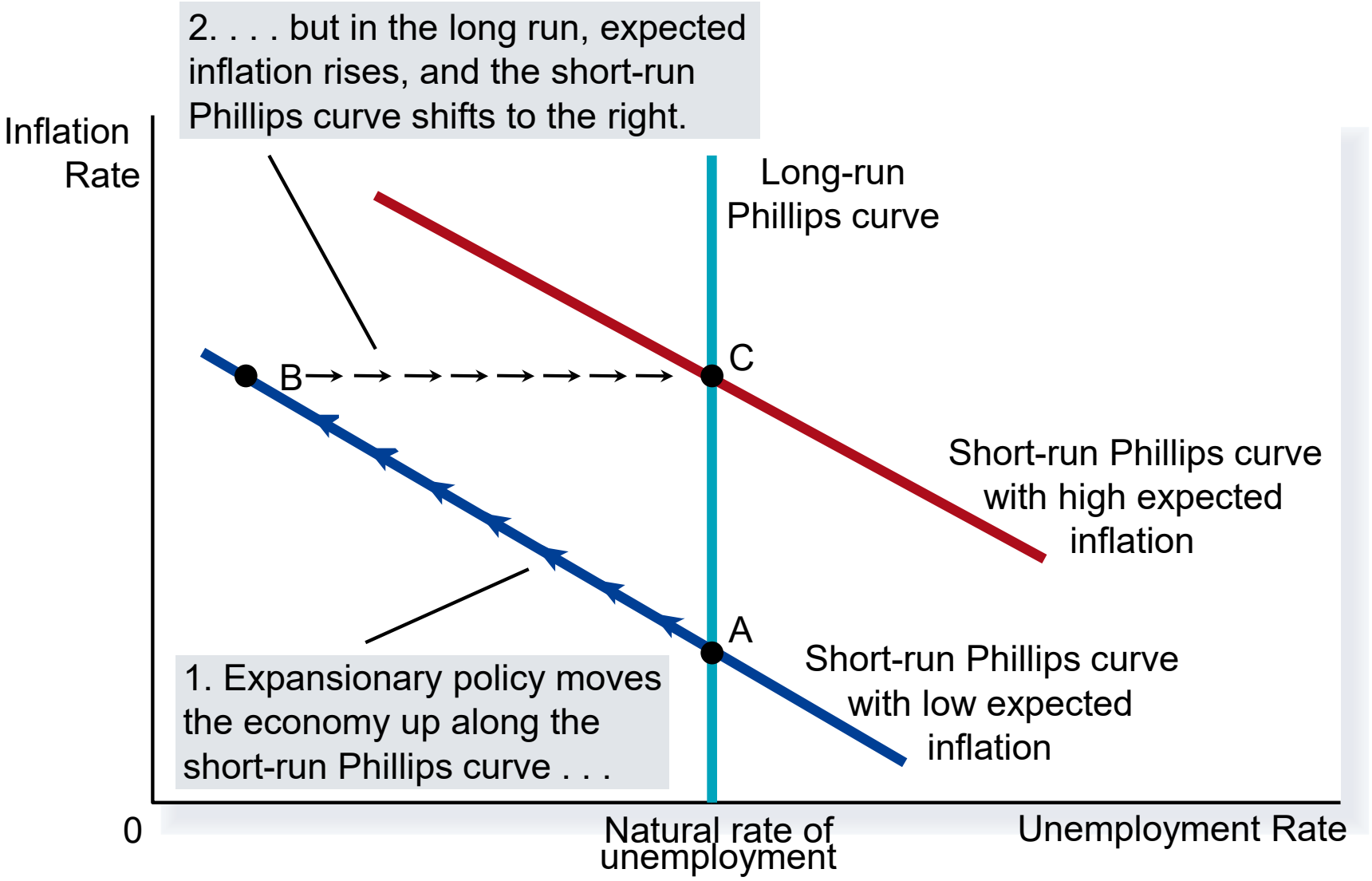
# The Short-Run Phillips Curve

- Expected inflation measures how much people expect the overall price level to change.
- Over time, expected inflation adjusts to changes in actual inflation.
  - The Central Bank's ability to create unexpected inflation exists only in the short run.
  - Once people anticipate inflation, the only way to get unemployment below the natural rate is for actual inflation to be above the anticipated rate.

$$\textit{Unemployment Rate} = \begin{matrix}\textit{Natural Rate}\\ \textit{of Unemployment}\end{matrix} - b \times \left(\begin{matrix}\textit{Actual}\\ \textit{Inflation}\end{matrix} - \begin{matrix}\textit{Expected}\\ \textit{Inflation}\end{matrix}\right)$$

Remember Short-run Aggregate Supply Curve: $Y = \hat{Y} + a \times (P - P^e)$

# How Expected Inflation Shifts the Short-Run Phillips Curve



# The Long-run Phillips Curve and Central Bank Independence

- The government might use an expansionary fiscal policy to boost aggregate demand with the intention of reducing unemployment below it natural rate
- If this is anticipated it will only increase inflation (move to point C)
- An alternative is to give the governor of the central bank control of monetary policy and tell her/him that she/he will be fired unless a specified low level of inflation is achieved over a reasonable time period
- What does this achieve?
- Without the expansionary policy the economy remains at point A
- The level of unemployment is no worse than what we ended up with following the government's previous attempt to reduce it by following an expansionary policy
- And inflation is lower
- So this is a better outcome (stay in point A)

# The Long-run Phillips Curve and Central Bank Independence

- We can also model this using game theory
- A government would prefer point B to point A
- Firms and workers know this and know that if they don’t raise their inflation expectations then the government will pursue an expansionary policy
- So firms and workers raise their inflation expectations and we move to point C
- Point C can be thought of as a Nash equilibrium
- But C is worse than A
- Hence the best thing for the government to do to maximize its chances of re-election is to make the central bank independent and hand over control of monetary policy to it

# The Long-run Phillips Curve and Central Bank Independence

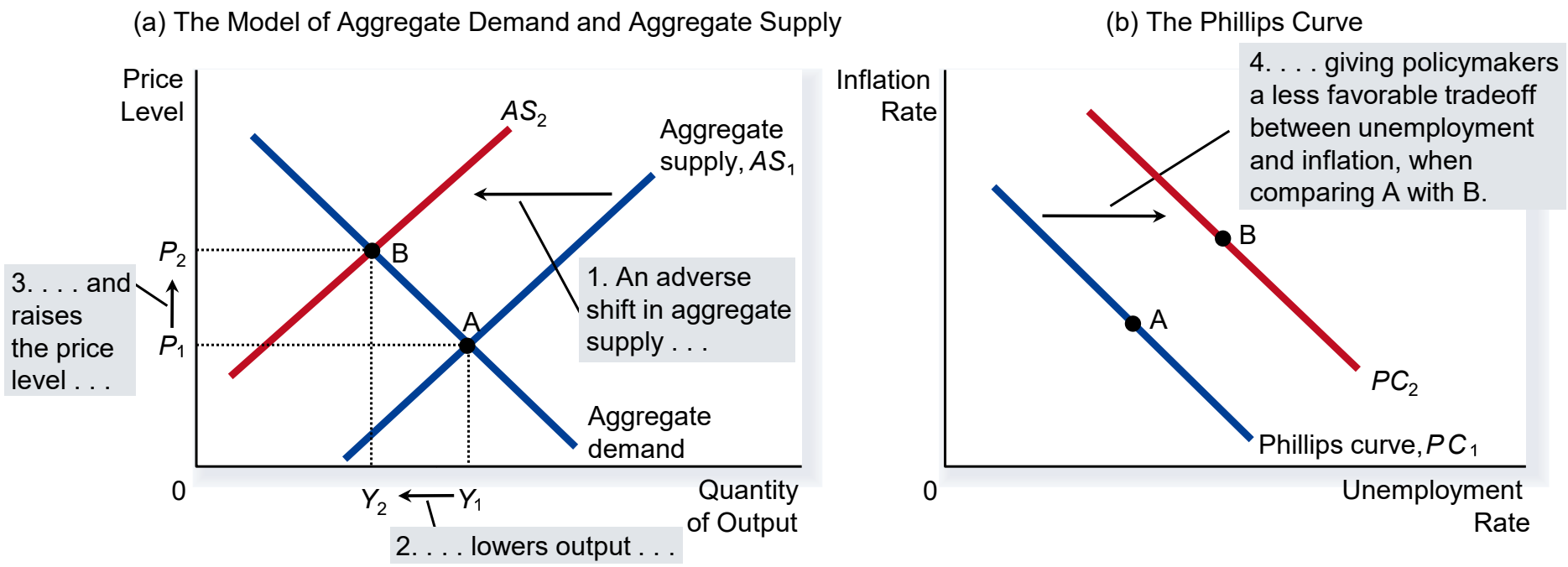
- The Bank of England was given independence and an inflation target in 1997
- The ECB has been independent since its inception in 1998 and is responsible for deciding what level of inflation should be targeted and then achieving that target level
- The US Federal Reserve is, like the ECB, independent in both design and implementation of monetary policy but does not pursue a formal inflation target
- The Swiss National Bank (SNB) obtained formal constitutional independence in May 2004, when the revised Article 99 of the Swiss Federal Constitution entered into force.
  - It became Switzerland's central bank in 1907 under the Federal Act on the Swiss National Bank
  - For most of its history it operated under the administration and supervision of the Confederation
  - Over the twentieth century, its de facto autonomy gradually increased, particularly as price stability became the primary monetary policy objective

# Shifts In The Phillips Curve: The Role Of Supply Shocks

- Historical events have shown that the short-run Phillips curve can shift due to changes in expectations
- The short-run Phillips curve also shifts because of shocks to aggregate supply
  - Adverse changes in aggregate supply can worsen the short-run trade-off between unemployment and inflation
  - An adverse supply shock gives policymakers a less favourable trade-off between inflation and unemployment
- A ***supply shock*** is an event that directly alters the firms' costs, and, as a result, the prices they charge
  - This shifts the economy's aggregate supply curve…
  - . . . and as a result, the Phillips curve

$$\textit{Unemployment Rate} = \begin{matrix}\textit{Natural Rate}\\ \textit{of Unemployment}\end{matrix} - b \times \left(\begin{matrix}\textit{Actual}\\ \textit{Inflation}\end{matrix} - \begin{matrix}\textit{Expected}\\ \textit{Inflation}\end{matrix}\right) + \begin{matrix}\textit{Supply}\\ \textit{Shock}\end{matrix}$$

# An Adverse Shock to Aggregate Supply



# Two causes of rising and falling inflation

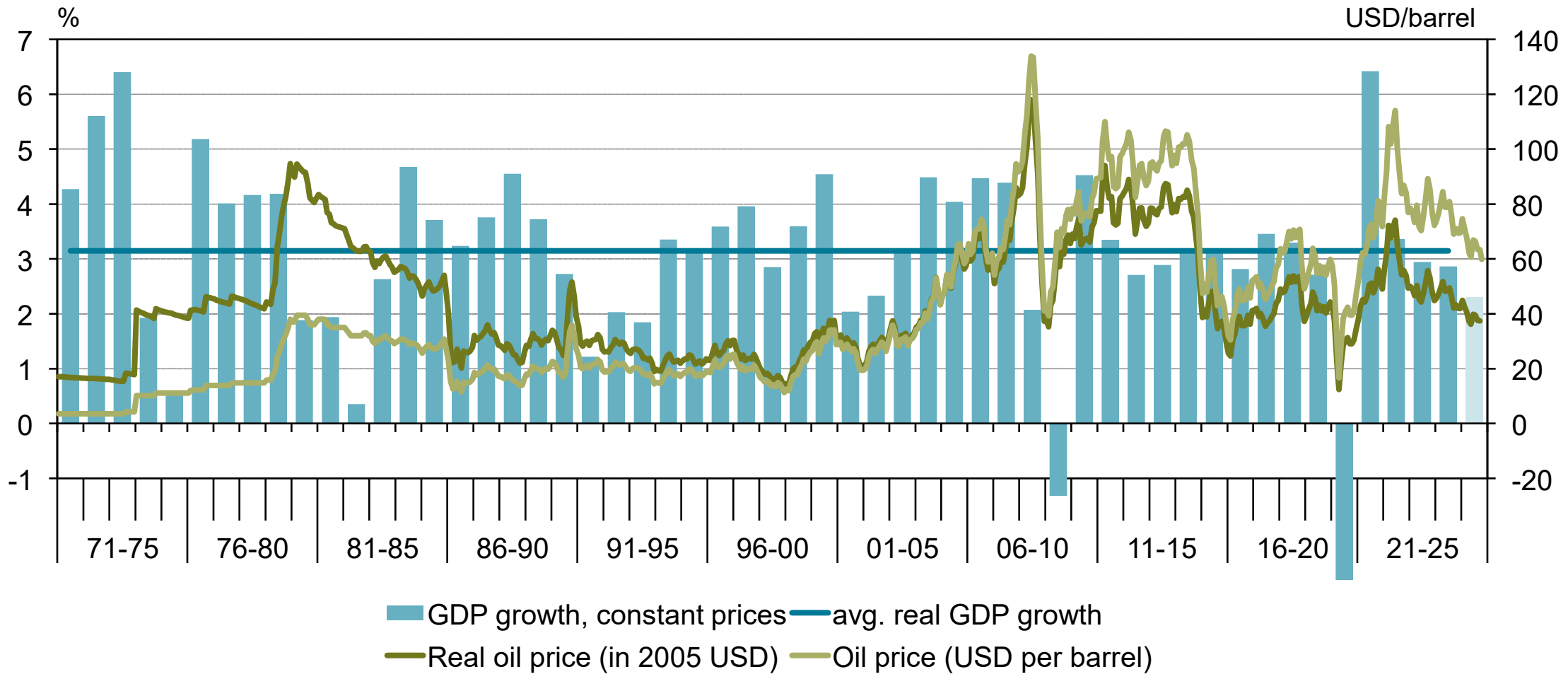
- Demand-pull inflation:
  - Inflation resulting from demand shocks
    - Positive shocks to aggregate demand cause unemployment to fall below its natural rate, which “pulls” the inflation rate up

- Cost-push inflation:
  - inflation resulting from supply shocks
    - Adverse supply shocks typically raise production costs and induce firms to raise prices, “pushing” inflation up

# Shifts in the Phillips Curve: The role of supply shocks

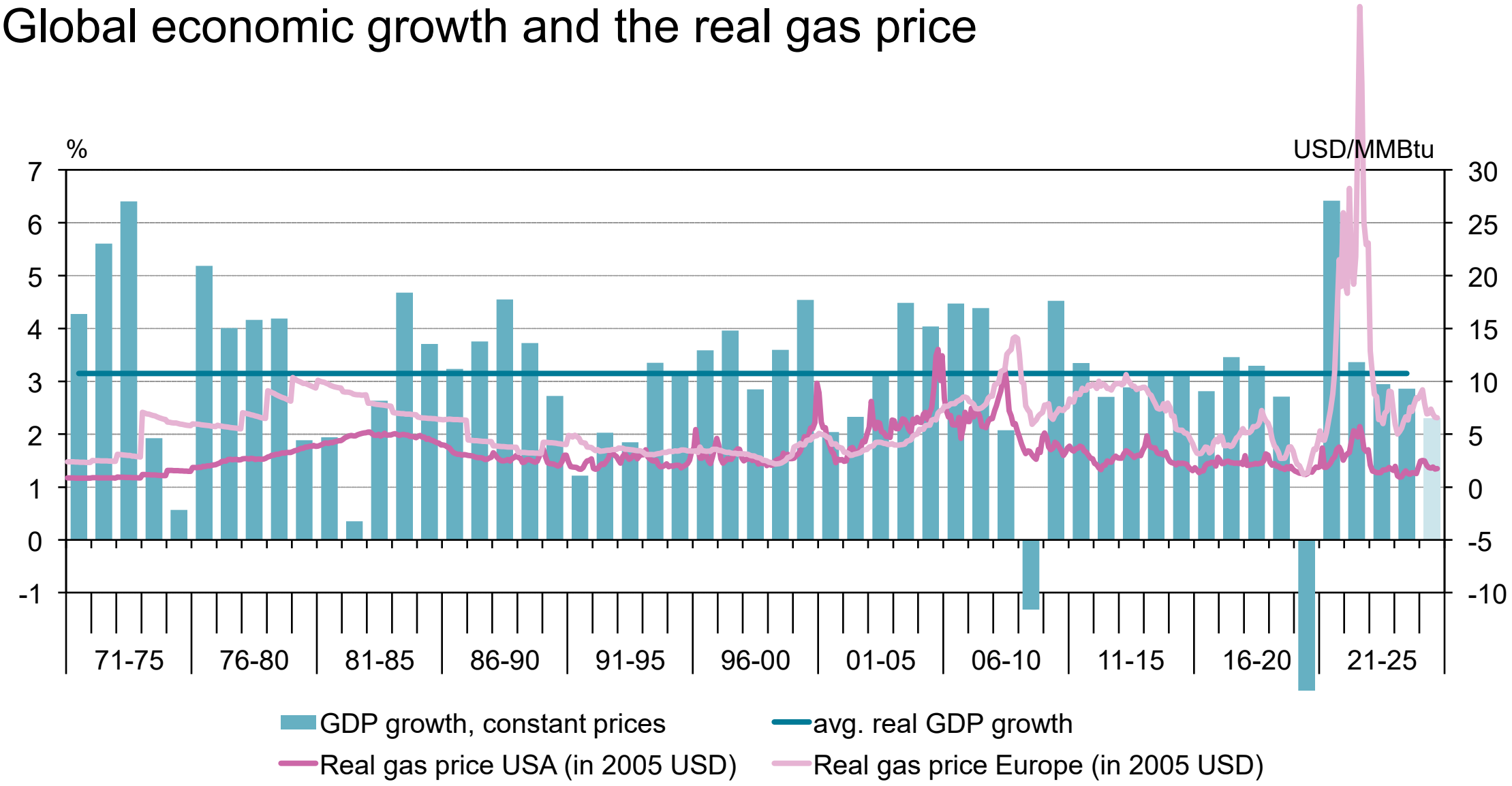
- In the 1970s, policymakers faced two choices when OPEC cut output and raised worldwide prices of petroleum
  - Fight the unemployment battle by expanding aggregate demand and accelerate inflation
  - Fight inflation by contracting aggregate demand and endure even higher unemployment

# Global economic growth and the oil price



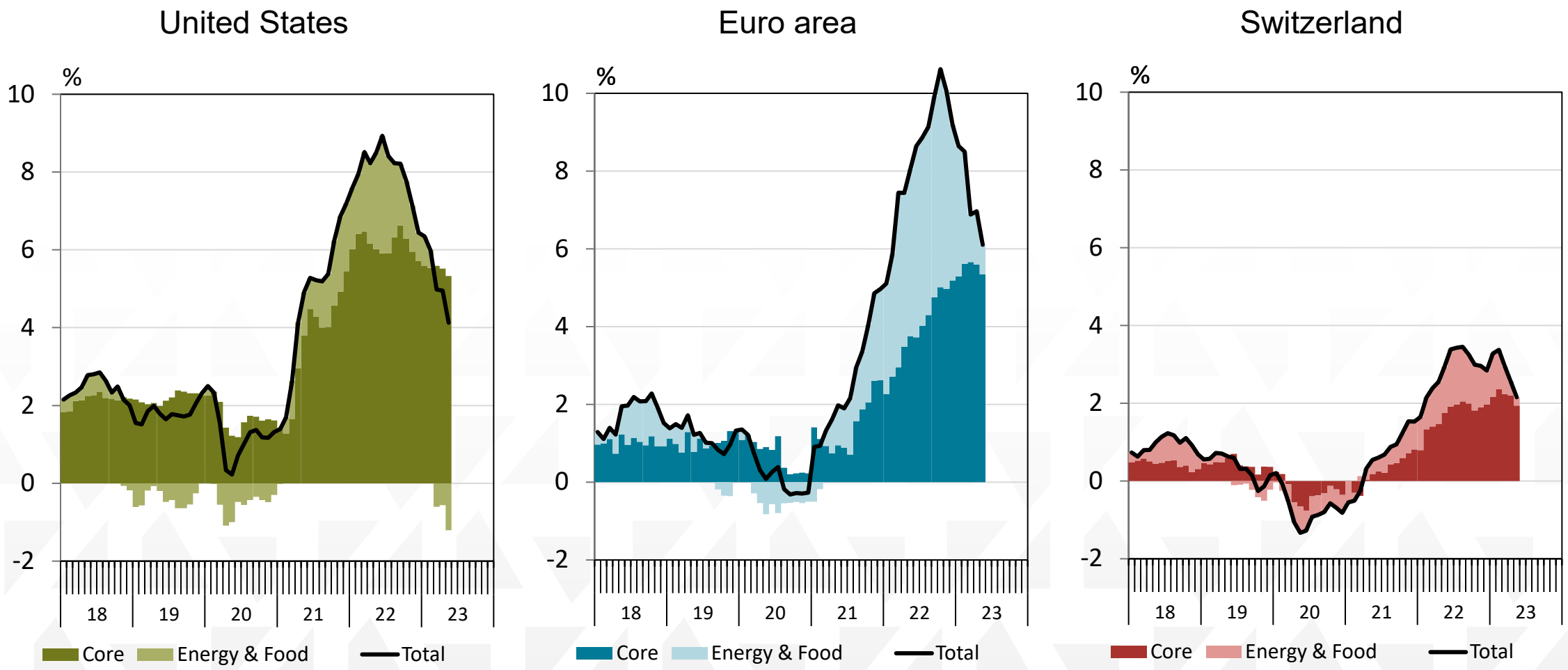

Sources: World Bank, ICE, US BLS

# Global economic growth and the real gas price



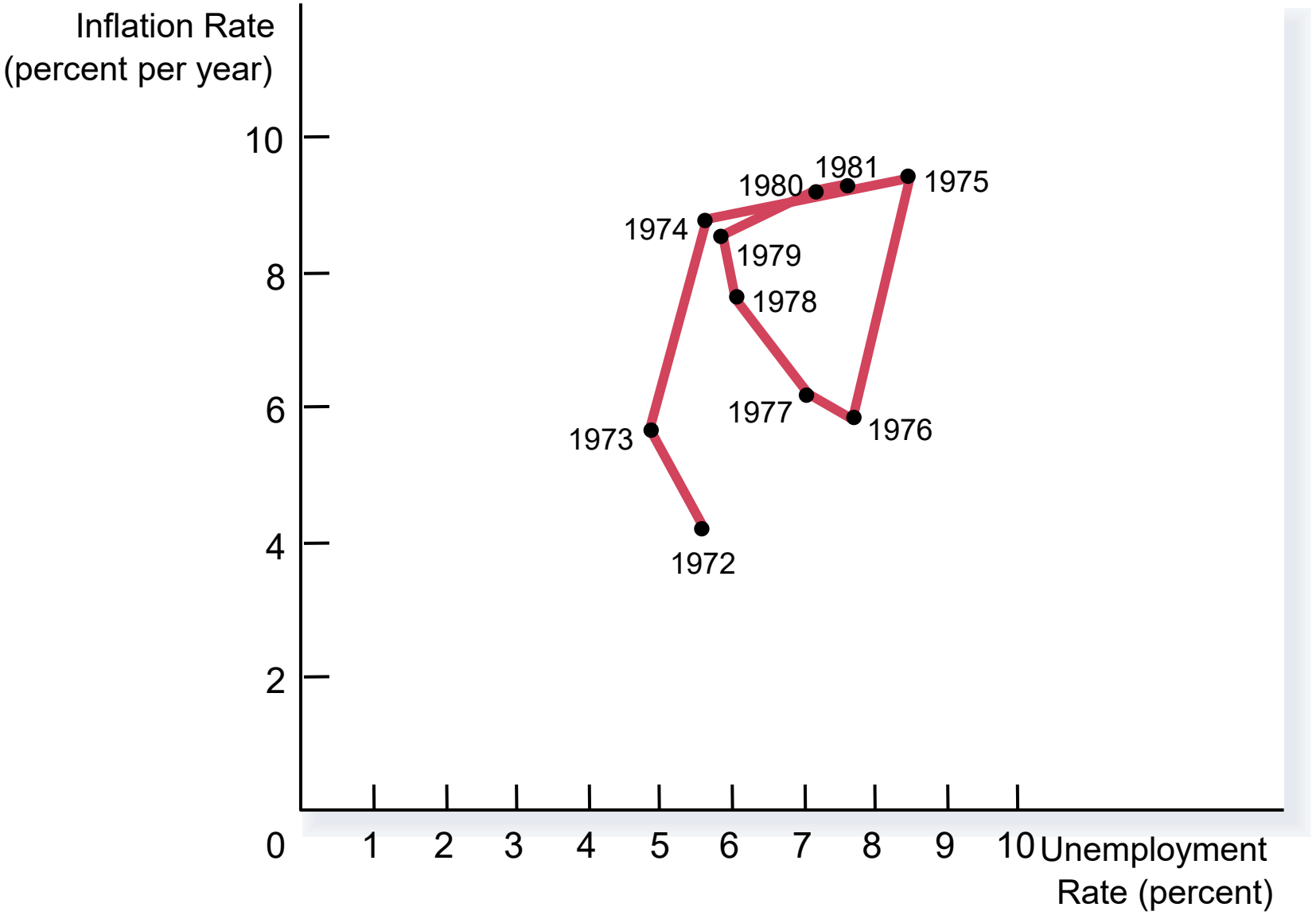

Sources: World Bank, US BLS

# Inflation rates broken down into core and energy & food

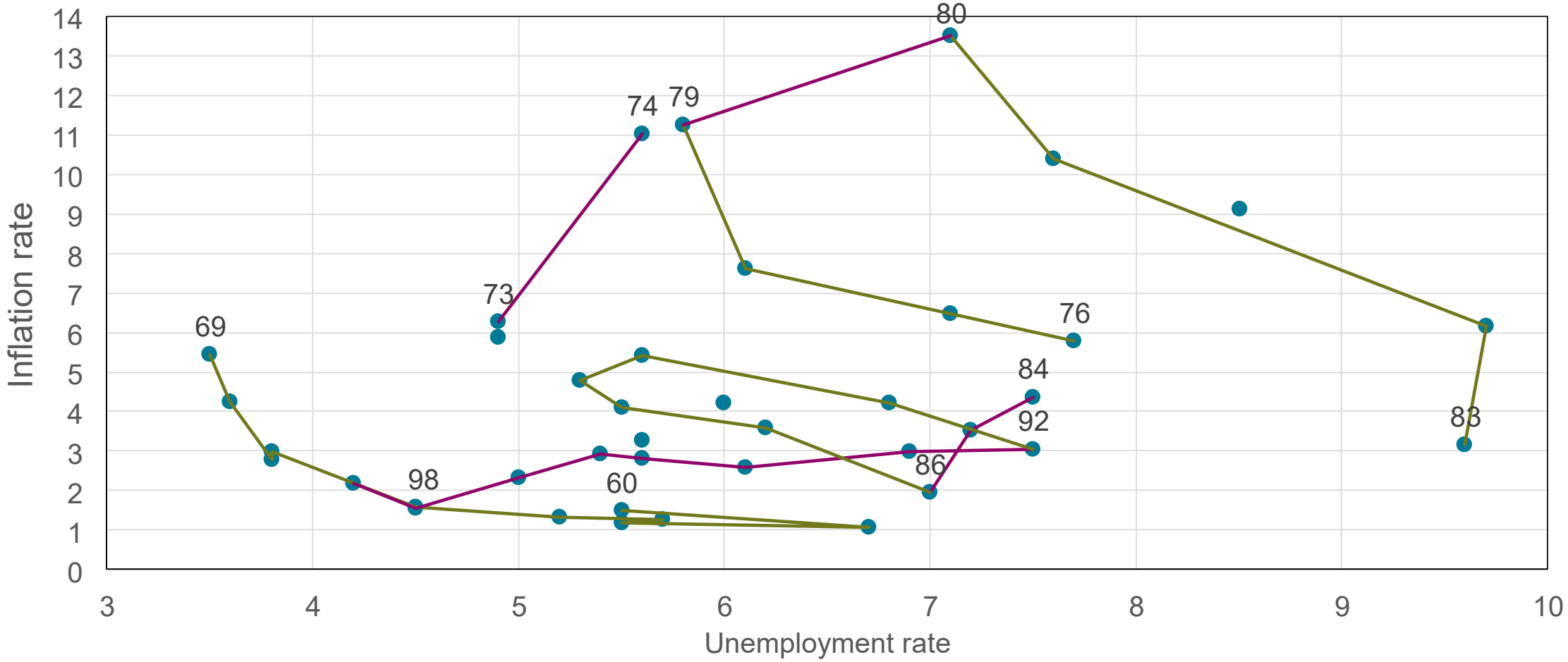



Sources: BLS, Eurostat, FSO

# The Supply Shocks of the 1970s



# Phillips Curve in the United States (1960-1999)

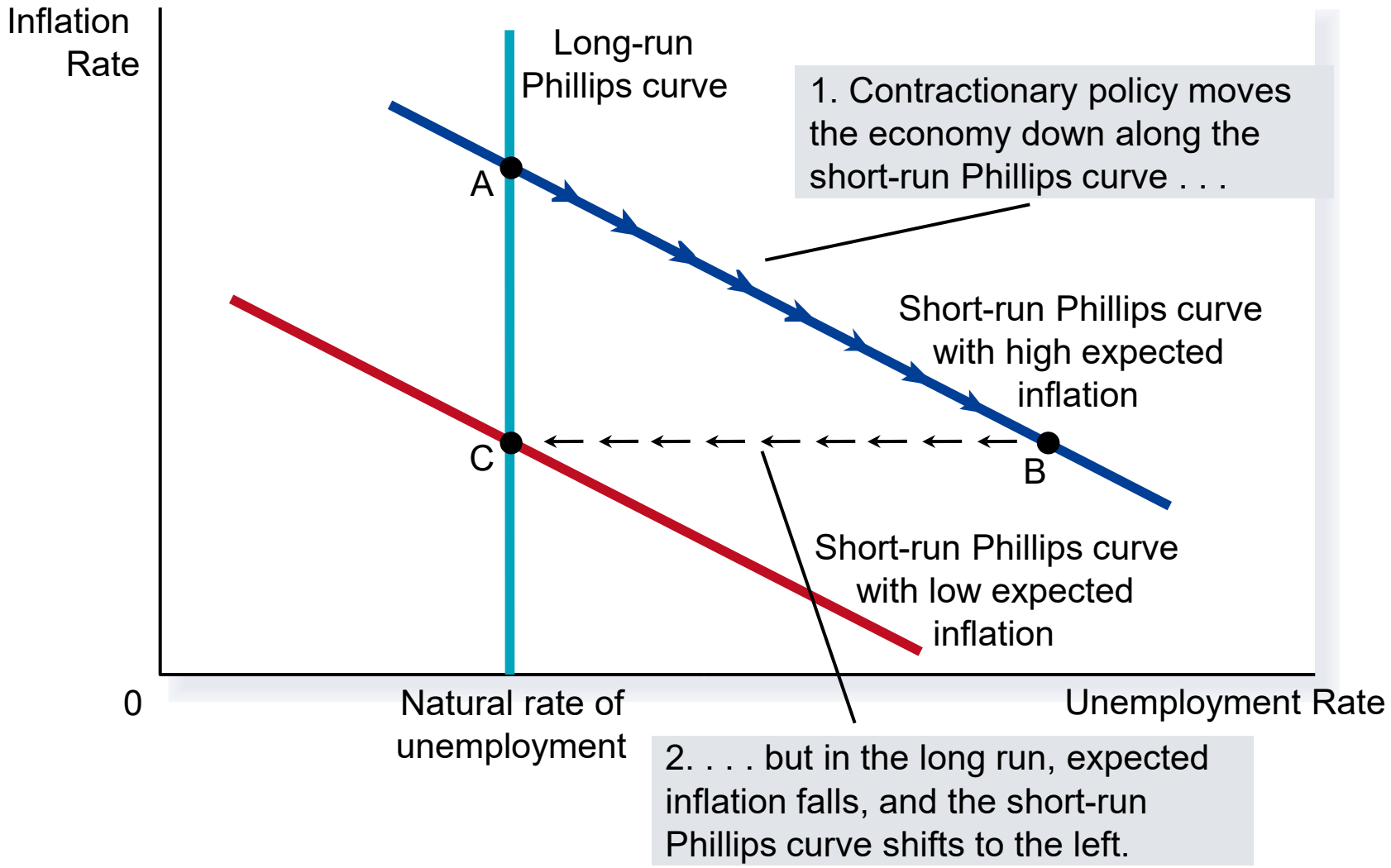



Sources: BEA, BLS

# The Cost of Reducing Inflation

- To reduce inflation, the CB has to pursue contractionary monetary policy
- When the CB slows the rate of money growth, it contracts aggregate demand
- This reduces the quantity of goods and services that firms produce
- This leads to a rise in unemployment

# Disinflationary Monetary Policy in the Short Run and the Long Run



# The Sacrifice Ratio

- To reduce inflation, an economy must endure a period of high unemployment and low output
- When the CB combats inflation, the economy moves down the short-run Phillips curve
- The economy experiences lower inflation, but at the cost of higher unemployment
- The ***sacrifice ratio*** is the number of percentage points of annual output that is lost in the process of reducing inflation by one percentage point
  - An estimate of the sacrifice ratio is five
  - To reduce inflation from about 10% to 4% in the US in 1979 would have required an estimated sacrifice of 30% of annual output!

# Rational Expectations and the Possibility of Costless Disinflation

- ***Adaptive expectations***:
  - People base their inflation expectations on recently observed inflation.

- ***Rational expectations***:
  - People base their inflation expectations on all available information, including information about current and prospective future policies
    - The theory of rational expectations suggests that people optimally use all the information they have, including information about government policies, when forecasting the future

- Expected inflation explains why there is a trade-off between inflation and unemployment in the short run but not in the long run
- How quickly the short-run trade-off disappears depends on how quickly expectations adjust
- The theory of rational expectations suggests that the sacrifice-ratio could be much smaller than estimated

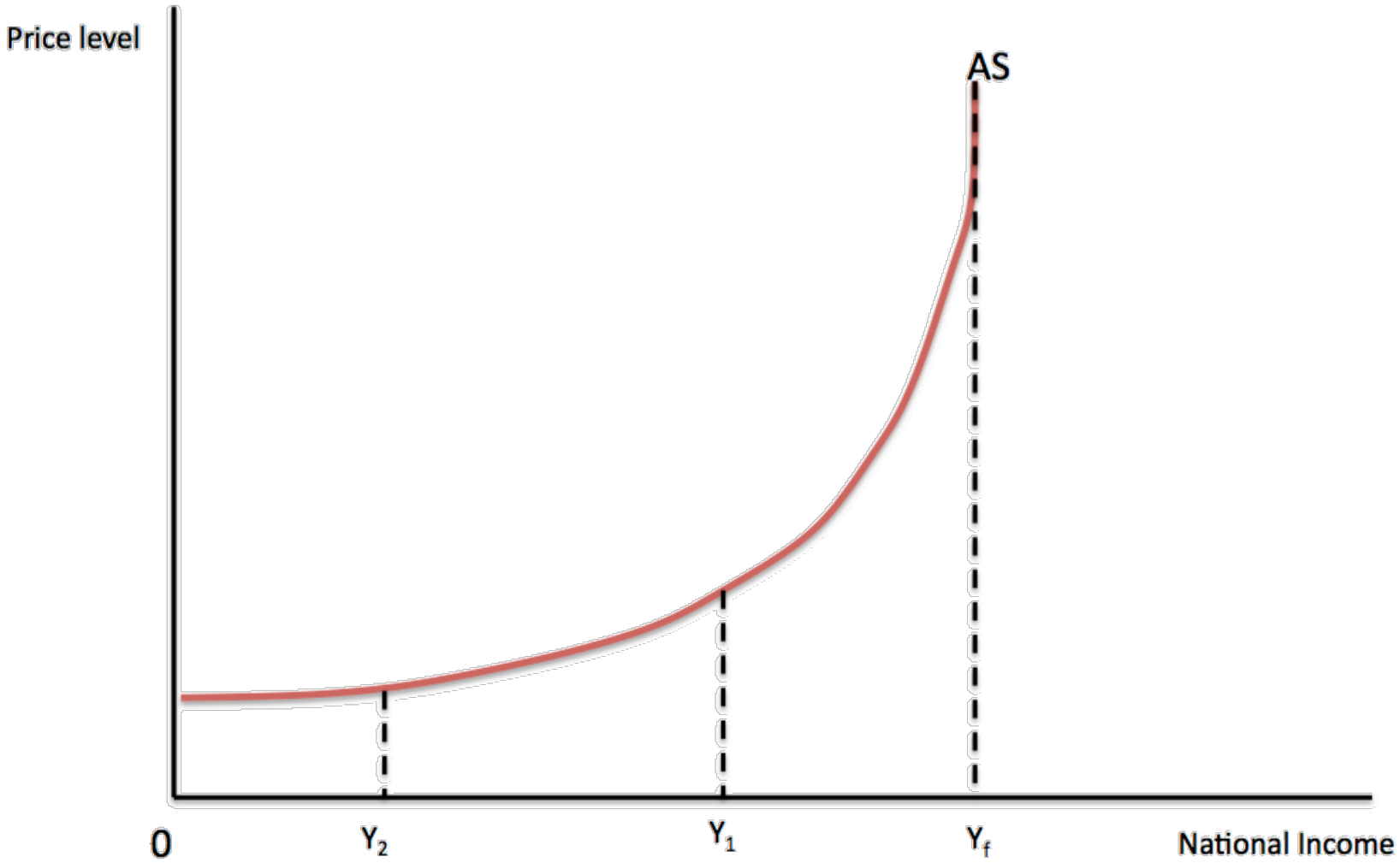
# The Volcker Disinflation

- When Paul Volcker was Fed chairman in the 1970s, inflation was widely viewed as one of the United States' foremost problems
- Volcker succeeded in reducing inflation (from 10 percent to 4 percent), but at the cost of high unemployment (about 10 percent in 1983)

# The Greenspan Era

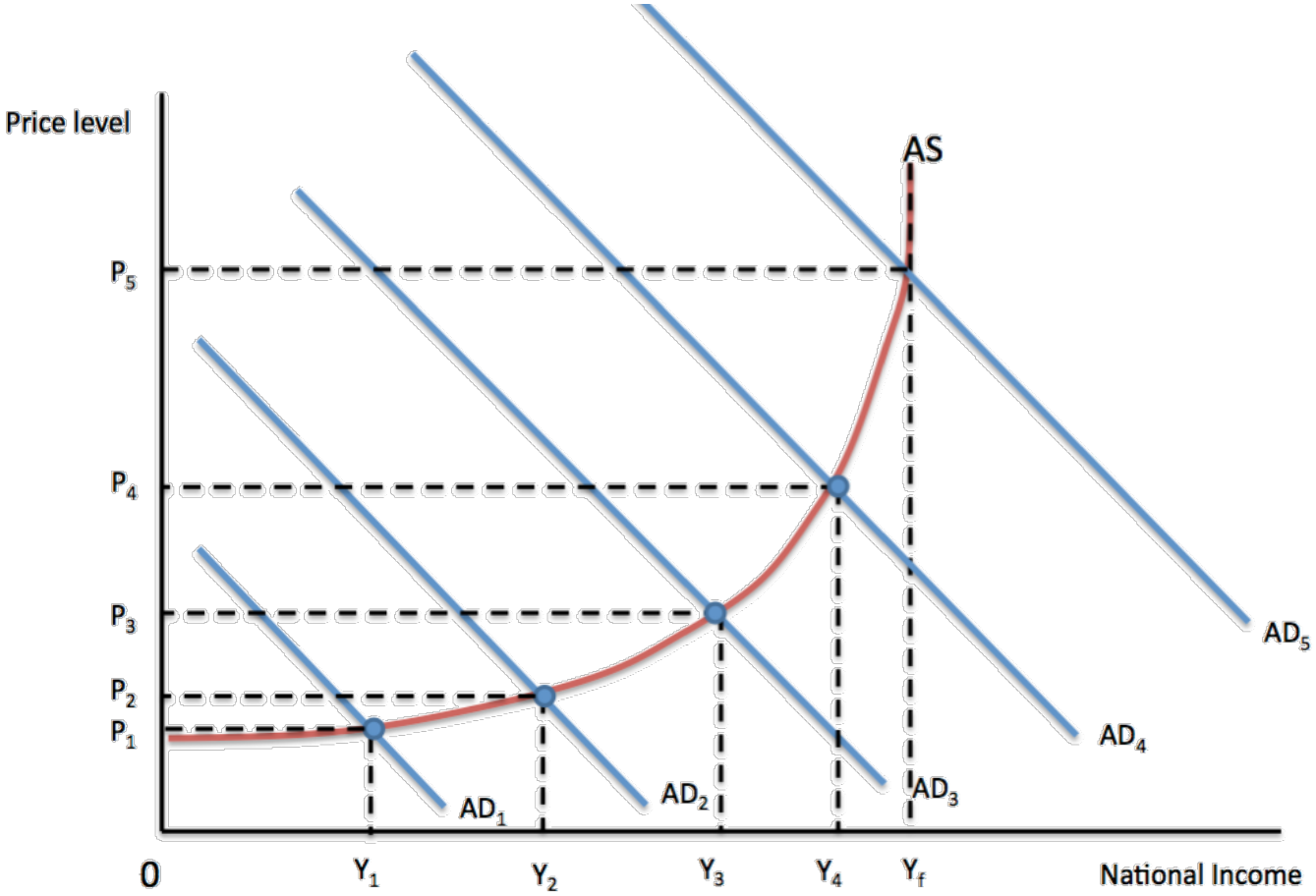
- Alan Greenspan's term as Fed chairman began with a favourable supply shock
  - In 1986, OPEC members abandoned their agreement to restrict supply
  - This led to falling inflation and falling unemployment

# The Shape of the New Keynesian Short-Run Aggregate Supply Curve



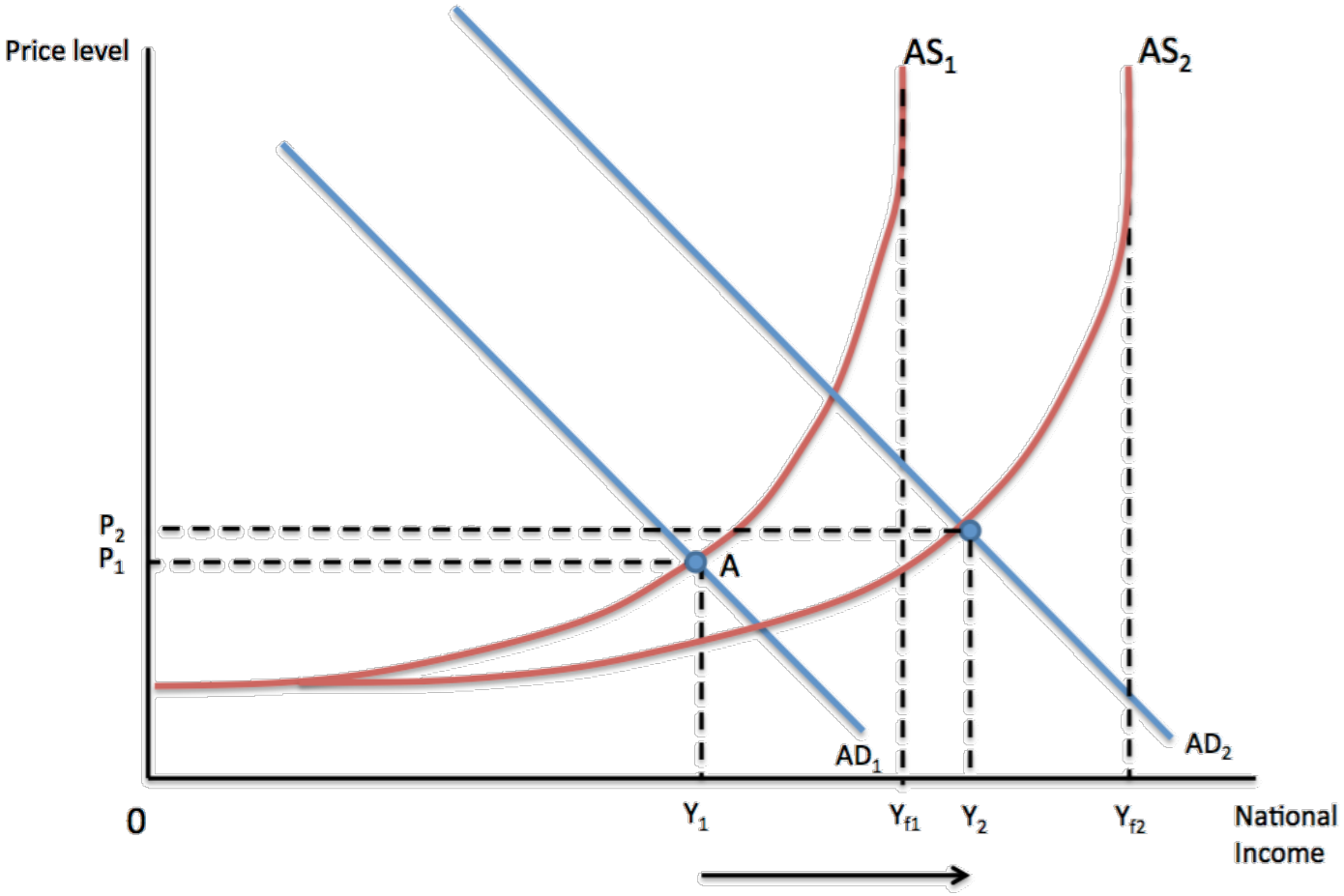
# The Aggregate Supply Curve and Production Bottlenecks

- The trade-off between an increase in national income and price rises depends on how large the output gap is
  - If AD continues to shift to the right eventually means that national income will only rise by relatively small amounts but the pressure on resources will cause the price level to rise more quickly
- Getting the output gap wrong when using fiscal tools can lead to unanticipated consequences
  - Rising prices with little growth
  - Large fall in national income with little impact on inflation levels



# Sustained Economic Growth

- Shifting the aggregate supply to the right can lead to sustained growth
  - Increases living standards
  - Keeps prices stable
- Supply-side economists improve the workings of free markets using
  1. ***Market-orientated*** supply-side policies designed to free up markets to improve resource allocation through more effective price signals
  2. ***Interventionist*** supply-side policies focused on improving the working of markets through investing in infrastructure, education and research and development



# 1. Market Orientated Supply-Side Policies

There are four elements to market-orientated supply-side policies.

A. Reform tax and welfare policy

B. Flexible labour markets

C. Reducing government spending

D. Privatization and deregulation

# 1.A. Reform tax and welfare policy

- Taxes are a disincentive to work and distort market outcomes
- Higher tax rates reduce the opportunity cost of leisure
- Higher taxes encourage tax avoidance and evasion
- Welfare payments act like a negative tax and therefore also distort market outcomes

- Cutting tax rates can (but do not have to) mean that tax revenues actually increase
  - Individual might work harder as his/her net return increases
  - It affects those looking for work
    - Raising the threshold for paying tax, increases the gap between welfare earnings and potential earnings
    - Those on benefits are less fussy about choosing jobs
    - Fewer would choose to be voluntarily unemployed

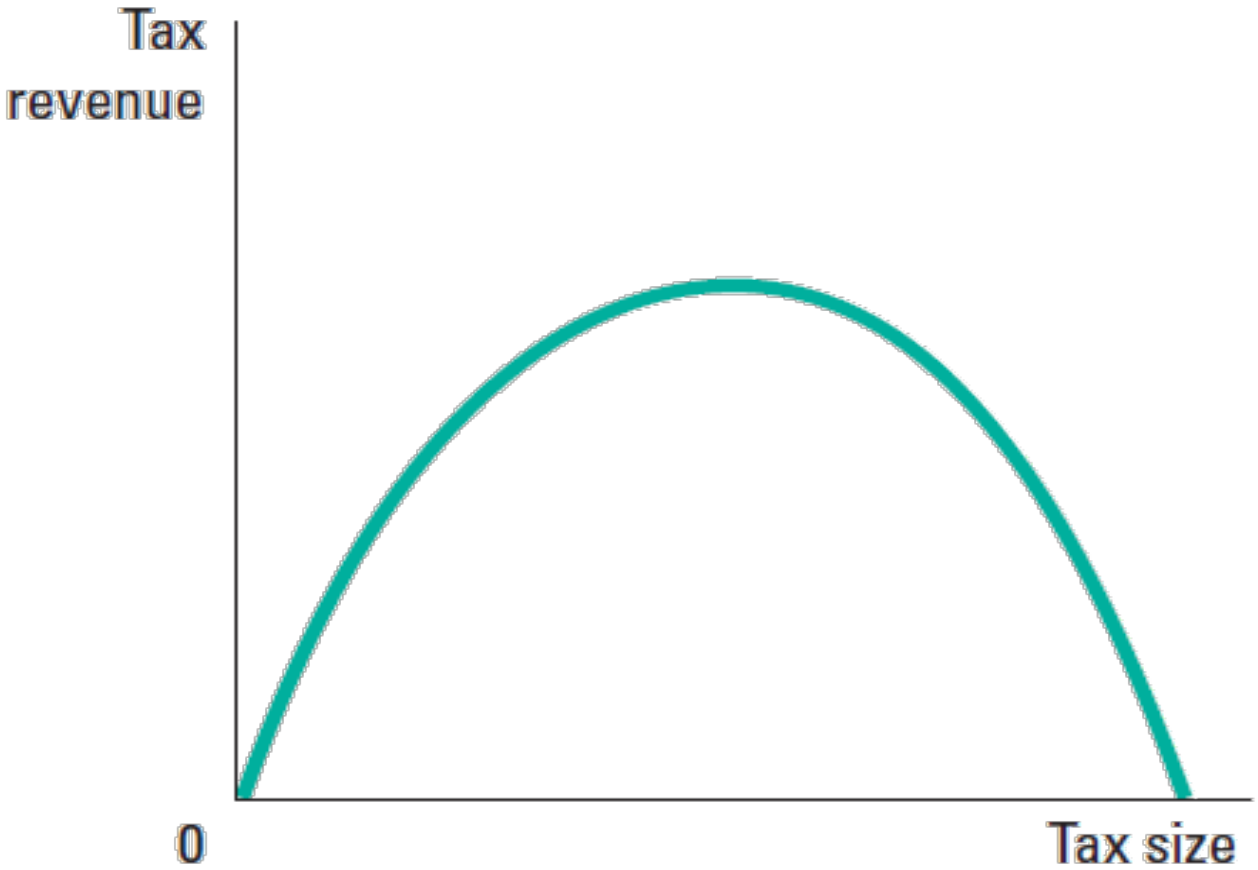
# Changes in Top Rates of Tax

**Top Rates of Income Tax in 9 European Counties, 1979–2012**

| Country | Top Income tax rate 1979 (%) | Top Income tax rate 1995 (%) | Top Income tax rate 2012 (%) |
|---|---|---|---|
| Belgium | 76.3 | 55.0 | 50.0 |
| Denmark | 66.0 | 63.5 | 55.0 |
| Germany | 56.0 | 53.0 | 45.0 |
| Ireland | 60.0 | 48.0 | 48.0 |
| Netherlands | 72.0 | 60.0 | 52.0 |
| Norway | 75.4 | 13.7 | 47.8 |
| Portugal | 80.0 | 40.0 | 46.5 |
| Sweden | 86.5 | 30.0 | 56.6 |
| UK | 83.0 | 40.0 | 50.0 |

# The Laffer Curve and Supply-Side Economics

- The ***Laffer curve*** looks at the relationship between tax rates and tax revenue
- Laffer argued that US taxes were too high and reducing them would increase tax revenue
- Cutting taxes to raise revenue may be correct if applied to taxpayers facing the highest tax rates, but most people face lower marginal rates
- The more elastic that supply and demand are in any market, the more taxes in that market distort behaviour, and the more likely it is that a tax cut will raise tax revenue



# 1.B. Flexible Labour Markets

- Three aspects to getting more flexible labour markets
  I. Reduce trade union power and encourage local level wage agreements
  II. Reduce rigidities in the labour market by increasing the ease of hiring and firing
     - Includes more short-term contracts
     - However, power balance may be too much with employers
  III. Improving market information for employers, workers and the unemployed

# 1.C. Reducing Government Spending

- The more money government spends the more it has to raise in tax revenue and/or borrow
  - Leads to crowding out effect
  - Government spending not as efficient as private sector investment
- If borrowing can be scaled back, interest rates will fall and this has an effect on firms' decisions to invest

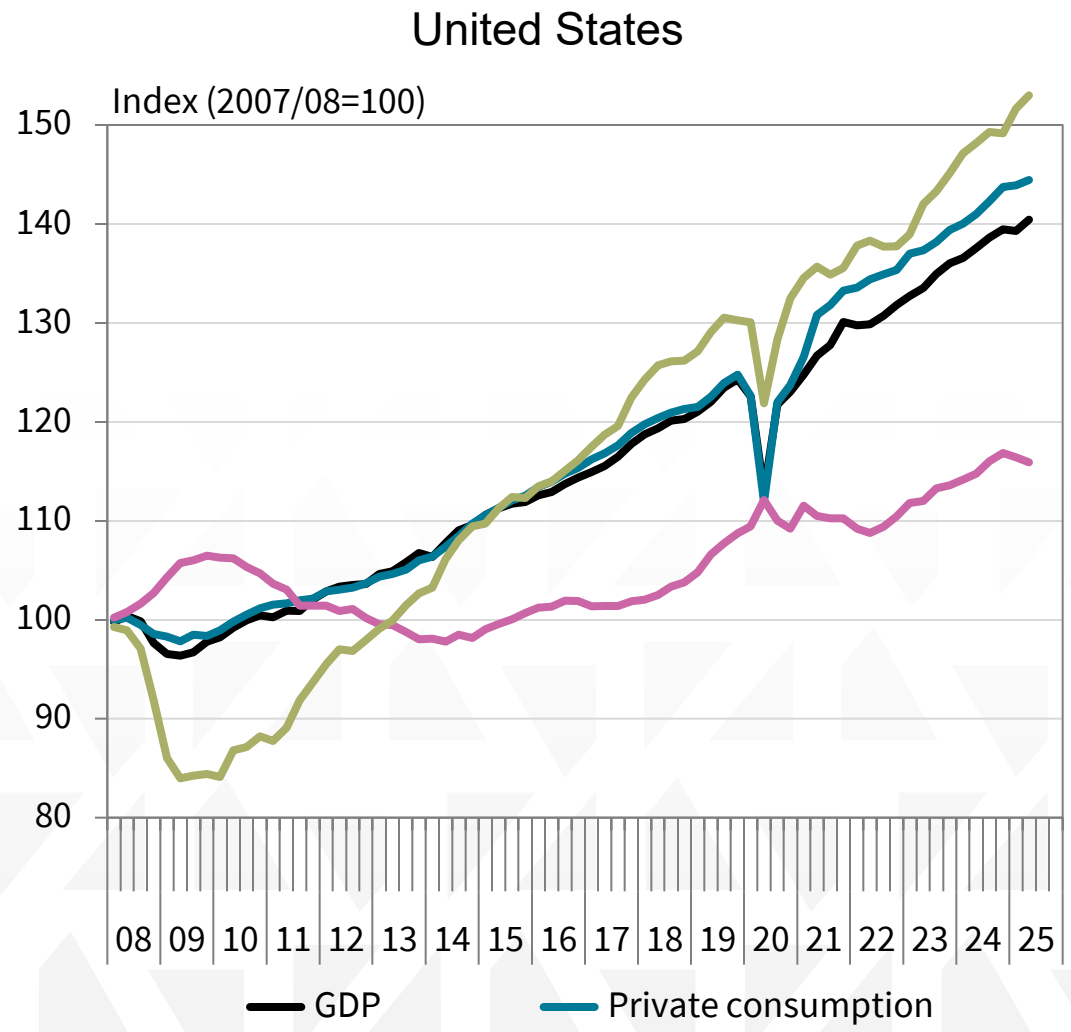
# 1.D. Privatization and Deregulation

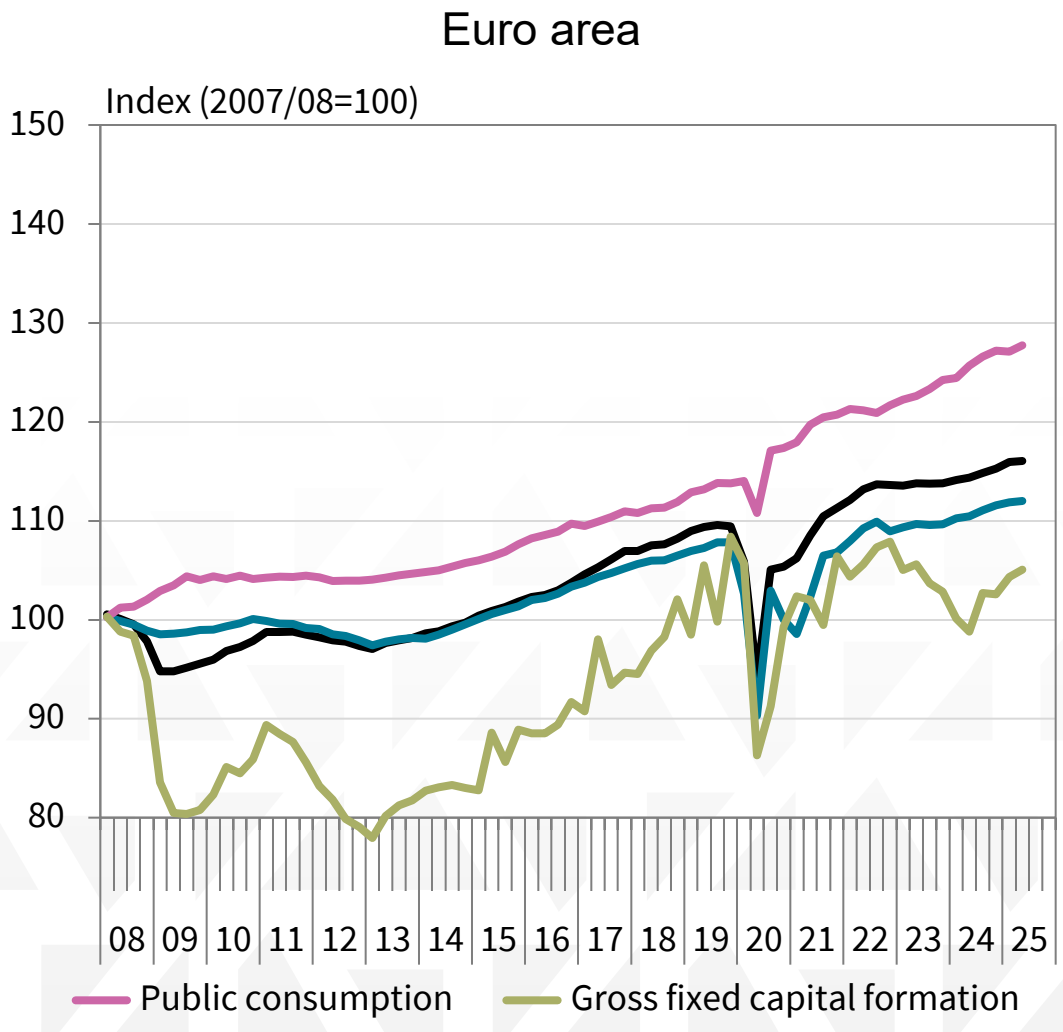
- ***Privatization*** is the transfer of public ownership of assets to the private sector
  - It is common in many European countries
  - It should lead to higher productivity and lower costs
  - However the privatization of monopolies with ineffective regulation can reduce consumer surplus
- ***Deregulation*** should remove some market imperfections
  - Deregulation went too far and appropriate checks and balances were not in place to prevent the financial crisis in 2007–2008

# 2. Interventionist Supply-Side Policies

- Four key aspects to interventionist supply-Side policies.
  I. Infrastructure Investment
     - Governments involved where there are long term benefits
  II. Investment in Education and Training
  III. Research and Development
     - Research bodies , tax relief or grants
  IV. Regional or Industrial Policies
     - Designed to encourage investment and the regional multiplier

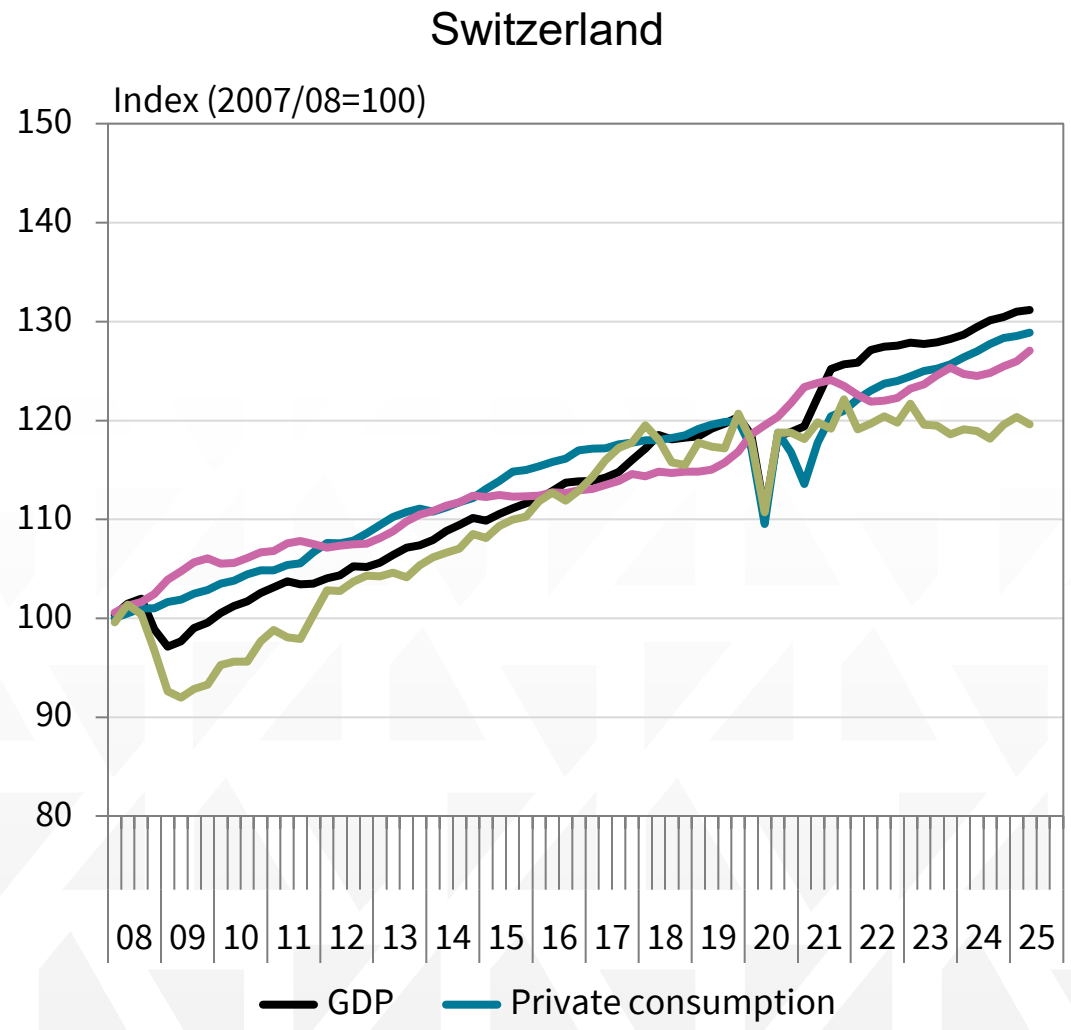
# GDP and its domestic spending components

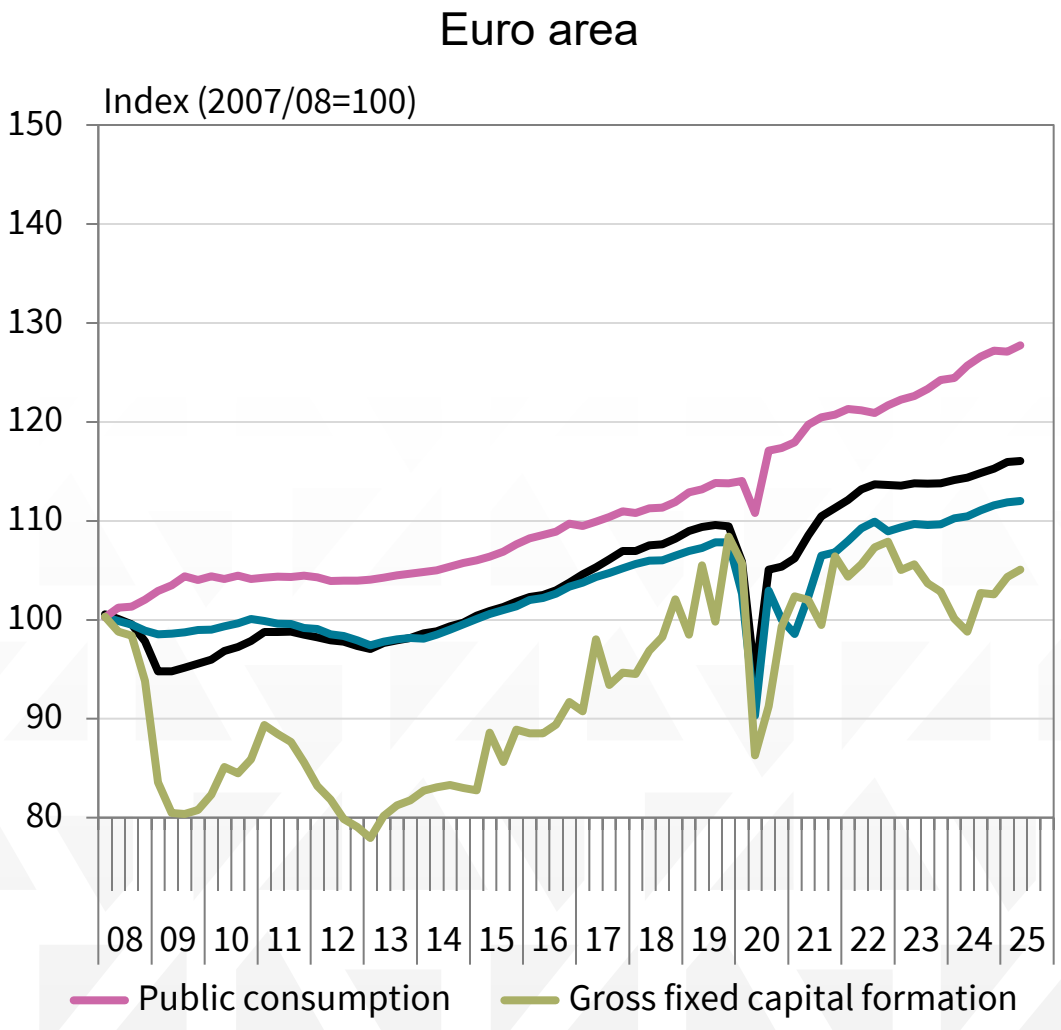



Sources: BEA, Eurostat

# GDP and its domestic spending components




Sources: BEA, Eurostat

# Summary

- The Phillips curve describes a negative relationship between inflation and unemployment
- By expanding aggregate demand, policymakers can choose a point on the Phillips curve with higher inflation and lower unemployment
- By contracting aggregate demand, policymakers can choose a point on the Phillips curve with lower inflation and higher unemployment
- The trade-off between inflation and unemployment described by the Phillips curve holds only in the short run
- The long-run Phillips curve is vertical at the natural rate of unemployment
- The short-run Phillips curve also shifts because of shocks to aggregate supply
- An adverse supply shock gives policymakers a less favorable trade-off between inflation and unemployment
- When the CB contracts growth in the money supply to reduce inflation, it moves the economy along the short-run Phillips curve

# Summary

- This results in temporarily high unemployment
- The cost of disinflation depends on how quickly expectations of inflation fall
- Supply-side policies became a focus of many governments in the 1980s and into the 1990s
- Bottlenecks can occur if trying to shift aggregate demand as factors of production become more scarce
- An adverse supply shock gives policymakers a less favourable trade-off between inflation and unemployment
- When the CB contracts growth in the money supply to reduce inflation, it moves the economy along the short-run Phillips curve
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